ContinualRAG-BEIR: Benchmarking and Diagnosing Bounded-Memory Eviction in Retrieval-Augmented Agents Under Sequential Domain Shift

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Abstract

When a retrieval-augmented generation (RAG) agent operates over a sequential stream of knowledge domains under a strict document memory budget, which documents should it retain and which should it evict? While recent continual learning work focuses either on parametric adapter tuning or unbounded memory graphs, production retrieval indices face a fundamental *non-parametric selection and eviction dilemma*: optimizing for repeated traffic (exploitation) versus preserving coverage for unseen novel queries (exploration). We introduce **ContinualRAG-BEIR**, a leak-free continual retrieval benchmark spanning up to 8 sequentially arriving BEIR domains (8,000 streaming documents, 40% memory cap) with disjoint online usage streams and held-out test queries, pre-registered exploratory/confirmatory seed governance, budget-matched result caching, and cross-embedder validity controls (`E5-small-v2` selection → `BGE-small-en-v1.5` retrieval). Through a pre-eviction predictability audit, we show that because test-relevant document reuse across novel queries is only 6.5%, policies driven purely by usage feedback (`LRU`, `LFU`, `DecayedHit`, `W-TinyLFU`, and global Gaussian Process utility models) gain +0.217 nDCG@10 on repeated queries ($p_{\text{Holm}} = 0.0039$) but fail to beat `Random` eviction on novel queries. Conversely, we discover a striking *Coreset Paradox*: classic diversity-maximizing coreset selection (`K-Center` and stratified `K-Center+Quota`) catastrophically degrades novel-query nDCG@10 by -0.041 to -0.063 below `Random` ($p \leq 0.0003$) because farthest-point traversal actively selects geometric *anti-hubs* (outlierness AUC = 0.401). Reversing this objective via training-free within-domain structural centrality (`HubDensity-Pure+Quota`) improves novel-query nDCG@10 by +0.027 to +0.042 over `Random` ($p_{\text{Holm}} \leq 0.0054$). Finally, we propose **CAGE** (*Centrality-Anchored Gated Eviction*), a dual-memory policy that pairs a multi-hit heavy-hitter admission gate ($\rho = 0.06$) with a domain-stratified hub-density core ($1 - \rho = 0.94$). Across 50 confirmatory evaluations spanning 5-domain and 8-domain streams and two embedding families (85 distinct confirmatory seeds overall), `CAGE+Quota` simultaneously beats `Random` on novel queries (+0.024 to +0.040 nDCG, $p_{\text{Holm}} \leq 0.0054$), beats `Density+Quota` on repeated queries (+0.114 to +0.120 nDCG, $p_{\text{Holm}} \leq 0.0002$), and is proven statistically non-inferior to pure structural eviction on novel queries within a 0.01 nDCG margin at 0.033–0.046 ms/doc ingestion latency.

1 Introduction

Retrieval-Augmented Generation (RAG) agents deployed in enterprise, scientific, and personal assistant workflows do not operate on static corpora. New domain corpora arrive sequentially as projects evolve, while hardware RAM, vector-index latency SLAs, and edge-device memory budgets impose strict upper bounds on the number of indexed passages $M_{\max}$. When a new domain t arrives and $|M_{t-1} \cup D_t| > M_{\max}$, the system must execute an eviction policy π to consolidate memory down to $M_{\max}$ documents.

Despite extensive literature on parametric continual learning (e.g., EWC, orthogonal LoRA, adapter merging) and static index pruning [8, 9], bounded-memory document eviction under sequential domain drift remains poorly standardized. Existing evaluations often suffer from three methodological flaws: (i) evaluating memory policies on the exact same query set used to drive online retention feedback (train/test leakage), (ii) evaluating selection and retrieval using a single embedding model without controlling for retriever self-selection circularity and BEIR pooling bias [1], and (iii) failing to compare smart vector eviction against a storage-matched exact-query result cache.

In this paper, we address these gaps by building **ContinualRAG-BEIR** and conducting a mechanistic investigation of bounded-memory retrieval across 26 eviction policies. Our contributions are fourfold:

1. **Leak-Free Continual Retrieval Benchmark & Governance:** We establish `ContinualRAG-BEIR` over 5-domain (`SciFact`, `NFCorpus`, `FiQA`, `ArguAna`, `SCIDOCS`) and 8-domain (+ `TREC-COVID`, `Quora`, `Touché-2020`) streams. Each domain's queries are

partitioned into a disjoint *Usage* pool (sampled via a Zipfian stream $p(q) \propto \text{rank}^{-s}$ for online feedback) and a held-out *Test* pool (novel queries). All claims are governed by a strict split between exploratory seeds (0–9) and locked confirmatory seed blocks (100–719) using Holm-Bonferroni corrected Wilcoxon signed-rank tests.

2. **Pre-Eviction Predictability Audit:** Before running any eviction policy, we audit the information-theoretic predictability of novel-query relevance. Across BEIR domains, only 6.5% of test-relevant documents are ever relevant to a usage query (and 0.0% in `FiQA` and `ArguAna`). Consequently, past usage relevance has an AUC of 0.396 (< 0.50) for predicting novel-query relevance, explaining why frequency/recency caches (`LRU`, `LFU`, `DecayedHit`, `W-TinyLFU`) and usage-trained Gaussian Processes (`GP-TS`) fail on novel queries. Conversely, training-free within-domain k-NN density achieves an AUC of 0.634 on unexposed documents.

3. **The Coreset Paradox & Anti-Hub Mechanism:** While coreset diversity (`K-Center`) is a standard baseline in continual learning [5], we prove that farthest-point coreset sampling catastrophically harms RAG retrieval (-0.041 to -0.063 nDCG below `Random`, $p \leq 0.0003$). High-dimensional embedding spaces exhibit *hubness* [2]: outliers have an AUC of 0.401 for novel relevance and behave as unretrievable *anti-hubs*. Prioritizing within-domain structural hubs (`Density+Quota`, `Hubness+Quota`, `HubDensity-Pure+Quota`) yields $+0.027$ to $+0.042$ nDCG gains over `Random` ($p_{\text{Holm}} \leq 0.0054$).

4. **Centrality-Anchored Gated Eviction (`CAGE+Quota`):** Naive dual-memory hybrids (`HitDensity+Quota` at $\rho = 0.25$) suffer a -0.017 to -0.020 novel-query penalty because 1-hit peripheral documents displace structural hubs. We introduce `CAGE+Quota`, which gates working-memory admission ($\rho = 0.06$) on multi-hit verification ($h_i \geq 1.2$) modulated by structural centrality, while allocating 94% of each domain's quota to a composite Hub-Density-Medoid core. `CAGE+Quota` is the only policy to simultaneously pass all confirmatory tests for novel superiority over `Random`, repeated superiority over `Density+Quota`, and formal 0.01-margin novel non-inferiority across all three benchmark regimes.

2 Related Work

Continual Learning in Retrieval and RAG. Recent work has begun studying continual adaptation in retrieval systems. HippoRAG 2 [3] models non-parametric continual memory via Personalized PageRank over knowledge graphs, while Hu et al. [4] propose Evolving Retrieval Memory (ERM) to drift document keys toward answered queries. Closer to streaming corpora, CREAM [5] adapts dense encoders using stratified coreset sampling, and Streaming RAG [6] applies heavy-hitter filtering for real-time knowledge bases. Unlike encoder-training approaches, our work isolates the fundamental *non-parametric bounded document eviction* problem under a fixed retriever and provides the first leak-free benchmark with cross-embedder controls.

Caching, Static Pruning, and Hubness. Classical caching algorithms such as TinyLFU and W-TinyLFU [7] combine recency admission windows with frequency sketches to maximize hit rates on skewed Zipfian streams. In information retrieval, static index pruning [8, 9] removes low-utility postings or passages prior to deployment. Our predictability audit bridges these two literatures with the geometry of high-dimensional spaces [2], showing that frequency sketches optimize repeated Zipfian traffic while intrinsic k-NN hubness/density priors prevent anti-hub retention on novel queries.

3 Problem Formulation & Benchmark Protocol

3.1 Sequential Bounded-Memory Retrieval

Let $\mathcal{T} = (1, 2, \ldots, T)$ denote a sequence of T arriving domains. At phase t, domain t arrives with a document set D_t ($|D_t| = N_d = 1{,}000$). The agent maintains a bounded memory $M_t \subseteq \bigcup_{j=1}^{t} D_j$ subject to $|M_t| \leq M_{\max}$. Upon arrival of D_t, the policy forms candidate pool $C_t = M_{t-1} \cup D_t$ and selects $M_t = \pi(C_t)$ if $|C_t| > M_{\max}$.

During phase t, the system serves an online usage stream U_t consisting of $N_{\text{use}} = 150$ queries per seen domain $j \leq t$, drawn i.i.d. from domain j's usage query pool Q_j^{use} according to a Zipf distribution $p_j(q_{(r)}) \propto r^{-s}$ ($s = 1.0$). For each query $q \in U_t$, the retriever returns the top-k ($k = 10$) documents from M_t by exact cosine similarity over L_2-normalized embeddings E^{ret}. The policy observes feedback in one of two modes: (1) `relevance` feedback (retrieved documents that are ground-truth positives for q, modeling click/thumbs-up signals) or (2) `retrieval` feedback (all top-k retrieved documents, requiring zero labels).

3.2 Evaluation Metrics and Leak-Free Split

Each domain j's queries are partitioned into Q_j^{use} (50%, observed via U_t) and a strictly disjoint held-out test pool Q_j^{test} (50%, never observed by any policy). After phase t, we evaluate:

- **Novel-Query Performance ($R_{t,j}^{\text{nov}}$):** Mean binary-relevance nDCG@10 on Q_j^{test} (where ideal DCG is

normalized over the full domain subsample D_j, not merely M_t). From the matrix $R^{\text{nov}} \in \mathbb{R}^{T\times T}$, we compute Final Novel nDCG ($\frac{1}{T}\sum_{j=1}^{T} R^{\text{nov}}_{T,j}$), Backward Transfer (BWT $= \frac{1}{T-1}\sum_{j=1}^{T-1}(R^{\text{nov}}_{T,j} - R^{\text{nov}}_{j,j})$), and Forgetting ($\frac{1}{T-1}\sum_{j=1}^{T-1}(\max_{l\in[j,T-1]} R^{\text{nov}}_{l,j} - R^{\text{nov}}_{T,j})$).

- **Repeated-Query Performance ($R^{\text{rep}}_{T,j}$):** Popularity-weighted nDCG@10 on Q^{use}_j under the final memory M_T, evaluated both on live memory alone (`final_rep`) and behind a first-served exact-query result cache (`final_rep_cached`).
- **Mixed Workload (Mixed(r)):** Mixed(r) $= r \cdot$ final_rep $+ (1-r) \cdot$ final_ndcg for repeat traffic share $r \in \{0.25, 0.50\}$.

3.3 Cross-Embedder Control & Statistical Governance

Because BEIR relevance judgments were originally pooled using existing retrieval systems [1], selecting documents with the same embedder used for evaluation could conflate genuine document utility with retriever-specific pooling bias. We therefore evaluate all policies under two regimes:

1. **Same-Embedder Regime (`BGE` → `BGE`):** Both policy selection (E^{sel}) and retrieval evaluation (E^{ret}) use `BAAI/bge-small-en-v1.5`.
2. **Cross-Embedder Control (`E5` → `BGE`):** Policies compute all geometric and query-proximity features using `intfloat/e5-small-v2` (E^{sel}), while retrieval and nDCG@10 scoring use `BAAI/bge-small-en-v1.5` (E^{ret}).

4 The Predictability Audit: Why Usage Feedback Fails on Novel Queries

Before comparing eviction algorithms, we ask a fundamental information-theoretic question: *Which observable signals carry mutual information with a document's relevance to unseen future queries?*

Table 1 reports the within-domain AUC of candidate signals for predicting binary relevance to held-out test queries (y_{test}). Two findings explain the behavior of all 26 policies:

1. **The 6.5% Reuse Bottleneck:** Across all 5 domains, only 6.5% of test-relevant documents are also relevant to any usage query (0.0% in `ArguAna` and `FiQA`). Because positive documents are partitioned across distinct queries, being relevant to a usage query actually has an AUC of **0.396** (< 0.50) for novel test queries! Consequently, any policy that prioritizes past usage hits (`LRU`, `LFU`, `DecayedHit`, `WindowHit`, or global `GP-TS` trained on hit rates) is fundamentally incapable of improving novel-query retrieval.
2. **The Anti-Hub Penalty vs. Local Centrality:** Geometric outlierness (−density, the implicit objective of `K-Center` diversity) has an AUC of **0.401** (and **0.404** under `E5-small-v2`). In contrast, simple training-free k-NN density ($k = 5$) achieves an AUC of **0.599** overall and **0.634** on documents not relevant to usage queries (**0.629** under `E5-small-v2`).

Table 1: Pre-eviction Predictability Audit across 5 BEIR domains (10 seeds, `BGE-small-en-v1.5`). Values report within-domain ROC-AUC for predicting relevance to held-out novel test queries ($y_{\text{test}} \in \{0,1\}$), alongside base and repeat rates.

Signal / Rate	**Argu.**	**FiQA**	**NFC.**	**SCID.**	**SciF.**	**ALL**
Base Rate $P(y_{\text{test}}=1)$	0.100	0.240	0.312	0.307	0.105	**0.213**
Repeat Rate	0.000	0.000	0.153	0.053	0.117	**0.065**
Usage-rel. (y_{use})	0.444	0.346	0.367	0.315	0.505	0.396
Max-sim usage Q	0.526	0.509	0.482	0.457	0.531	0.501
Top-3 sim usage Q	0.525	0.530	0.490	0.495	0.544	0.517
Outlierness (−dens)	0.347	0.381	0.472	0.378	0.428	0.401
kNN Density (k=5)	**0.653**	**0.619**	**0.528**	**0.622**	**0.572**	**0.599**
Density $\mid \neg y_{\text{use}}$	**0.672**	**0.670**	**0.545**	**0.689**	**0.592**	**0.634**

5 Methodology: From Anti-Hub Avoidance to CAGE

5.1 Why Coreset Diversity (`K-Center`) and Un-Quota'd Priors Fail

In high-dimensional L_2-normalized embedding spaces, cosine similarity distributions exhibit skewness known as *hubness* [2]: a subset of central documents (hubs) appear in the k-nearest neighbor lists of many queries and documents, whereas peripheral points (*anti-hubs*) almost never appear in any top-k list. Greedy farthest-point traversal (`K-Center`) explicitly selects points that minimize maximum similarity to already chosen points, filling the memory budget with extreme anti-hubs. Furthermore, global ranking without domain quotas (`Density`) fails under `BGE-small-en-v1.5` (0.2771 vs. 0.3185 for `Density+Quota`) because average pairwise cosine similarity varies across domains, causing tighter domains to monopolize memory (see Figure 2).

5.2 Proposed Method: Centrality-Anchored Gated Eviction (`CAGE+Quota`)

To achieve Pareto optimality across both novel and repeated queries without incurring $O(B \cdot N^2)$ submodular greedy overhead, we design **CAGE** (`CAGE+Quota`, Algorithm 1):

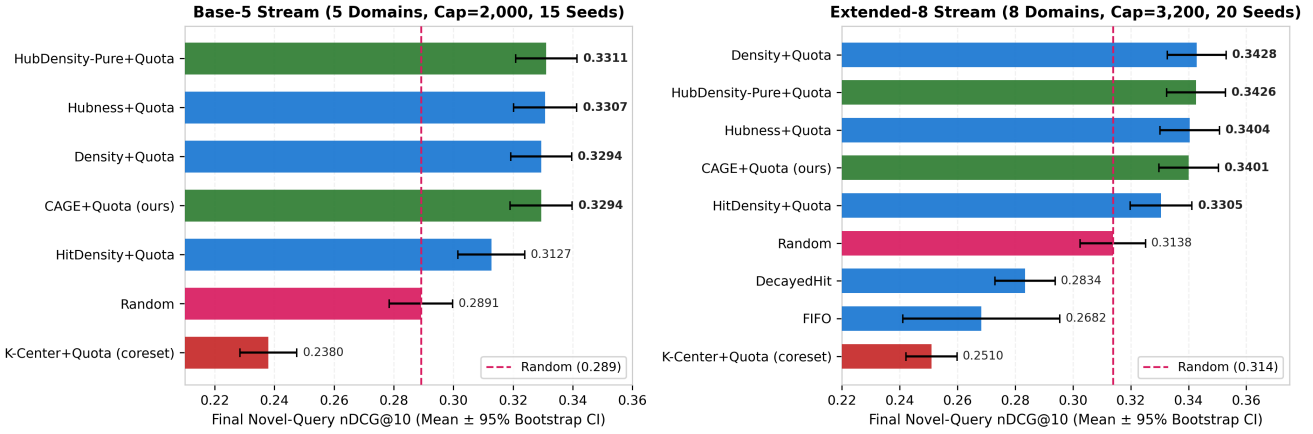


Figure 1: Final Novel-Query nDCG@10 (mean and 95% bootstrap CIs) across policies on Base-5 and Extended-8 streams (`E5-small-v2` selection → `BGE-small-en-v1.5` retrieval). Domain-stratified structural hub priors (`HubDensity-Pure+Quota`, `Hubness+Quota`, `Density+Quota`) and our gated dual-memory policy (`CAGE+Quota`) significantly outperform `Random`, whereas coreset diversity (`K-Center`) degrades retrieval.

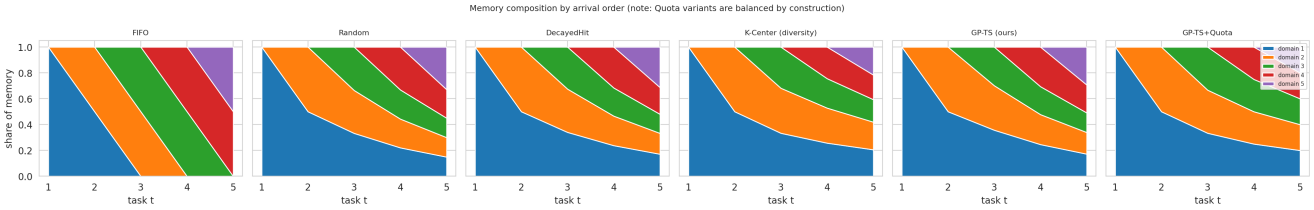


Figure 2: Memory composition by domain arrival order across phases $t = 1 \dots 5$. Un-quota'd temporal policies (`FIFO`) completely overwrite early domains, while un-quota'd global policies (`DecayedHit`, `K-Center`, `GP-TS`) suffer from cross-domain scale imbalance. Quota-enforced policies maintain exact domain parity.

1. Composite Structural Centrality (C_i): For each domain d present in candidate pool C_t with domain subset $C_{t,d}$ ($n_d = |C_{t,d}|$), we compute three training-free geometric statistics over the selection embeddings $E^{\text{sel}}_{C_{t,d}}$:

$$\text{Dens}(i) = \frac{1}{k_1} \sum_{j \in \mathcal{N}_{k_1}(i)} \langle e_i, e_j \rangle, \quad (k_1 = 5) \tag{1}$$

$$\text{Hub}(i) = \sum_{j \in C_{t,d} \setminus \{i\}} \mathbb{I}\big(i \in \mathcal{N}_{k_2}(j)\big), \quad (k_2 = 10) \tag{2}$$

$$\text{Med}(i) = \left\langle e_i, \frac{\mu_d}{\|\mu_d\|_2} \right\rangle, \quad \mu_d = \frac{1}{n_d} \sum_{j \in C_{t,d}} e_j \tag{3}$$

where $\mathcal{N}_k(i)$ denotes the top-k nearest neighbors of i within $C_{t,d}$. Let $R(x)_i = \frac{\text{rank}(x_i)}{n_d} \in (0, 1]$ denote within-domain percentile rank normalization. The composite structural centrality score is:

$$C_i = 0.45\, R(\text{Dens})_i + 0.45\, R(\text{Hub})_i + 0.10\, R(\text{Med})_i \tag{4}$$

Setting the working-memory ratio $\rho = 0$ yields `HubDensity-Pure+Quota`, a pure structural policy that allocates $B_d = \lfloor M_{\max}/|\mathcal{D}_t| \rfloor$ slots per seen domain to the documents with highest C_i.

2. Multi-Hit Centrality-Modulated Admission Gate ($\rho = 0.06$): In hybrid policies such as `HitDensity+Quota` ($\rho = 0.25$), reserving 25% of memory for any document with decayed hits $h_i > 0$ admits noisy 1-hit anti-hubs and starves novel-query coverage (failing non-inferiority by -0.017 to -0.020 nDCG). Under a Zipf ($s = 1.0$) query stream, repeated traffic is heavily concentrated in the top 5–6% of documents. `CAGE+Quota` therefore enforces a strict **Multi-Hit Admission Gate** within each domain d:

$$\mathcal{H}_d = \{ i \in C_{t,d} \mid h_i \geq \tau_{\text{hit}} \}, \quad (\tau_{\text{hit}} = 1.2) \tag{5}$$

Qualifying multi-hit documents $i \in \mathcal{H}_d$ are ranked by a **centrality-modulated hit score** $S^{\text{hot}}_i = \hat{r}_i \cdot (0.5 + 0.5\, C_i)$, and at most $B^{\text{hot}}_d = \min(|\mathcal{H}_d|, \lfloor \rho B_d \rfloor)$ ($\rho = 0.06$, i.e., at most 24 slots per domain when $B_d = 400$) are pinned. The remaining $B_d - B^{\text{hot}}_d$ slots (94–100% of the domain quota) are filled by the highest-centrality documents C_i.

Proposition 1 (Bounded Novel-Query Regret of `CAGE+Quota`). *Because `CAGE+Quota` restricts working-memory overrides to at most $\rho = 0.06$ of each domain's quota B_d and weights candidate overrides by $(0.5 + 0.5\, C_i)$, at least $1 - \rho = 94\%$ of retained documents in every domain are identical to the top-$(1 - \rho)$ structural hubs*

Algorithm 1 `CAGE+Quota` Consolidation Policy

Require: Candidate pool C_t, memory cap $M_{\max}$, active domains $\mathcal{D}_t$, decayed hits h, hit rates $\hat{r}$, quota ratio $\rho = 0.06$, gate $\tau_{\text{hit}} = 1.2$.

$B_d \leftarrow \lfloor M_{\max}/|\mathcal{D}_t| \rfloor, \quad M_t \leftarrow \emptyset$
for each domain $d \in \mathcal{D}_t$ with candidates $C_{t,d}$ **do**
  Compute $C_i \leftarrow 0.45R(\text{Dens})_i + 0.45R(\text{Hub})_i + 0.10R(\text{Med})_i$ for $i \in C_{t,d}$ using $E^{\text{sel}}_{C_{t,d}}$
  $\mathcal{H}_d \leftarrow \{i \in C_{t,d} \mid h_i \geq \tau_{\text{hit}}\}$
  $S^{\text{hot}}_i \leftarrow \hat{r}_i \cdot (0.5 + 0.5\, C_i)$ for $i \in \mathcal{H}_d$
  $P^{\text{hot}}_d \leftarrow \text{Top-}k(\mathcal{H}_d, S^{\text{hot}}, k = \min(|\mathcal{H}_d|, \lfloor \rho B_d \rfloor))$
  $P^{\text{core}}_d \leftarrow \text{Top-}k(C_{t,d} \setminus P^{\text{hot}}_d, C, k = B_d - |P^{\text{hot}}_d|)$
  $M_t \leftarrow M_t \cup P^{\text{hot}}_d \cup P^{\text{core}}_d$
end for
return M_t (filling any remainder $M_{\max} - |M_t|$ by top C_i)

of ***HubDensity-Pure+Quota****, upper-bounding worst-case novel-query nDCG displacement while capturing the Zipfian head of repeated queries.*

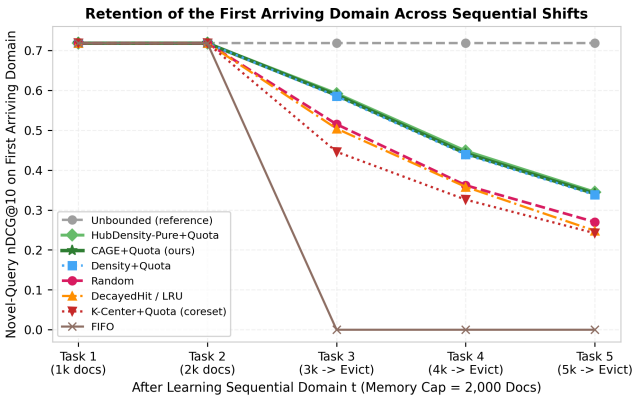


Figure 3: Retention of the first arriving domain's Novel nDCG@10 across sequential task arrivals $t = 1 \ldots 5$. Memory pressure begins at $t = 3$ (3,000 seen documents vs. cap $M_{\max} = 2{,}000$). `FIFO` completely forgets Task 1 by $t = 4$, and `K-Center` suffers severe anti-hub degradation, while `CAGE+Quota` and `HubDensity-Pure+Quota` preserve the highest retention.

6 Experimental Results

6.1 Main Exploratory Findings (Seeds 0–9)

Table 2 and Figure 1 summarize the exploratory evaluation across policies on `Base-5`. First, all pure usage-feedback policies (`DecayedHit`, `LRU`, `LFU`, `WindowHit`) achieve massive gains on repeated queries (0.5180–0.5770 vs. 0.2900 for `Random`) but score below `Random` on novel queries (0.2570–0.2824 vs. 0.2998). Second, `GP-TS (ours)` in its original global 8-PCA form scores 0.2956 (Regime 1) and 0.3017 (Regime 2), failing to beat `Random` ($p = 0.6953$) because 8 global PCA dimensions compress cross-domain shifts rather than within-domain relevance. Upgrading the GP to rank within each domain using $d = 24$ SVD features plus local density (`GP-WithinDom+Quota`) improves performance to 0.3220 novel nDCG and 0.4630 repeated nDCG, confirming our diagnosis. Third, training-free geometric hub priors (`Hubness+Quota`, `Density+Quota`) top all baselines on novel queries (0.3314 and 0.3313 under `E5` → `BGE`) at 0.02–0.03 ms/doc. Figure 3 illustrates the retention trajectory of the first arriving task as subsequent domains force document evictions.

6.2 Pre-Registered Confirmatory Tests (Seeds 100–719)

To eliminate multiple-testing bias, all key findings were tested once on disjoint confirmatory seed blocks (85 unique confirmatory seeds across Blocks 1–7):

- **Confirmatory Block 1 (Seeds 100–109):** `H1` (`Density+Quota` > `Random` on novel nDCG) is confirmed in both Regime 1 (+0.0180, 95% CI $[+0.0071, +0.0296]$, $p_{\text{Holm}} = 0.0137$) and Regime 2 (+0.0326, 95% CI $[+0.0197, +0.0452]$, $p_{\text{Holm}} = 0.0039$). `H2` (`DecayedHit` > `Random` on repeated nDCG) is confirmed (+0.2170, $p_{\text{Holm}} = 0.0039$). `H3` (equivalence of `DecayedHit` and `Random` on novel queries within ±0.02) is rejected (95% CI $[-0.0283, +0.0043]$), demonstrating that pure hit-rate eviction risks novel-query degradation.
- **Confirmatory Block 2 (Seeds 200–209):** `H4` (`Hubness+Quota` > `Random` on novel nDCG) is confirmed in Regime 1 (+0.0331, $p_{\text{Holm}} = 0.0039$) and Regime 2 (+0.0474, $p_{\text{Holm}} = 0.0039$). `H5` (`Hubness+Quota` ≠ `Density+Quota`) is not significant ($p_{\text{Holm}} = 0.6953$ and 0.3223), showing both capture the same anti-hub avoidance mechanism.
- **Confirmatory Block 3 (Seeds 300–309):** `HitDensity+Quota` ($\rho = 0.25$) confirms `H6b` (beating `Density+Quota` on repeated queries by +0.1431, $p_{\text{Holm}} = 0.0039$), but fails `H6a` ($p_{\text{Holm}} = 0.3750$ and 0.0840) and fails `H6c` non-inferiority on novel queries (−0.0196 and −0.0167, violating the −0.01 margin) due to un-gated 1-hit slot starvation.
- **Confirmatory Blocks 4 & 5 (Submodular Coverage Ablations, Seeds 400–509):** On Seeds 400–409, dense submodular facility location (`SubmodFL-Pure+Quota`) ties `Density+Quota` (0.3329 vs. 0.3330, $p = 0.6406$) because raw cosine similarities saturate after one medoid pick, while `GraphPPR+Quota` beats `Random` by +0.0291 ($p_{\text{Holm}} = 0.0293$). On Seeds 500–509, enforcing cluster-spreading via whitened sparse-kNN facility location (`SubmodSparse-Pure+Quota`) beats `Random` (+0.0236 to +0.0304, $p_{\text{Holm}} \leq 0.0273$) but under-

Table 2: Exploratory Benchmark Results on **Base-5** (5 BEIR domains, 5,000 streaming docs, memory cap $M_{\max} = 2{,}000$, 10 exploratory seeds 0–9, relevance feedback) across Same-Embedder (`BGE` → `BGE`) and Cross-Embedder Control (`E5` → `BGE`) regimes.

	Regime 1: Same-Embedder (`BGE` → `BGE`)						Regime 2: Cross-Embedder Control (`E5` → `BGE`)					
Strategy	**Novel (95% CI)**	**Rep.**	**Cache**	**BWT**	**Onl.**	**ms/doc**	**Novel (95% CI)**	**Rep.**	**Cache**	**BWT**	**Onl.**	**ms/doc**
Oracle (test labels)	0.665 [0.657, 0.673]	—	—	+0.025	0.864	0.0002	0.665 [0.657, 0.673]	—	—	+0.025	0.864	0.0002
Unbounded (100%)	0.617 [0.610, 0.624]	0.619	0.622	-0.005	0.842	0.0001	0.617 [0.610, 0.624]	0.619	0.622	-0.005	0.842	0.0001
SubmodFL-Pure+Quota	—	—	—	—	—	—	**0.332** [0.321, 0.343]	0.304	0.469	-0.208	0.583	1.7596
Hubness+Quota	**0.322** [0.313, 0.330]	0.298	0.460	-0.209	0.573	0.0207	**0.331** [0.319, 0.342]	0.326	0.467	-0.200	0.590	0.0269
Density+Quota	**0.319** [0.311, 0.326]	0.294	0.469	-0.222	0.571	0.0143	**0.331** [0.320, 0.342]	0.303	0.468	-0.208	0.582	0.0196
GraphPPR+Quota	—	—	—	—	—	—	0.323 [0.313, 0.334]	0.327	0.469	-0.204	0.598	0.0962
GP-WithinDom+Quota	—	—	—	—	—	—	0.322 [0.311, 0.331]	0.463	0.467	-0.220	0.716	0.5104
SubmodFL+Quota	—	—	—	—	—	—	0.318 [0.306, 0.331]	0.464	0.446	-0.193	0.703	1.4094
HitDensity+Quota	0.303 [0.290, 0.315]	0.461	0.447	-0.210	0.709	0.0140	0.317 [0.305, 0.330]	0.466	0.448	-0.195	0.703	0.0175
GP-TS+Quota	0.298 [0.288, 0.307]	0.352	0.462	-0.219	0.605	0.3171	0.311 [0.297, 0.324]	0.351	0.466	-0.209	0.617	0.3751
GP-TS (global PCA=8)	0.296 [0.286, 0.305]	0.351	0.487	-0.266	0.601	0.2738	0.302 [0.291, 0.312]	0.362	0.493	-0.256	0.624	0.3147
Random	0.300 [0.287, 0.313]	0.290	0.519	-0.282	0.559	0.0002	0.300 [0.287, 0.313]	0.290	0.519	-0.282	0.559	0.0002
Quota-Random	0.292 [0.277, 0.306]	0.292	0.465	-0.225	0.565	0.0005	0.292 [0.277, 0.306]	0.292	0.465	-0.225	0.565	0.0006
Reservoir	0.288 [0.280, 0.296]	0.275	0.463	-0.234	0.556	0.0039	0.288 [0.280, 0.296]	0.275	0.463	-0.234	0.556	0.0046
DecayedHit / LRU / LFU	0.282 [0.273, 0.292]	**0.518**	0.488	-0.297	0.754	0.0002	0.282 [0.273, 0.292]	**0.518**	0.488	-0.297	0.754	0.0003
DecayedHit+Quota	0.279 [0.269, 0.290]	0.477	0.453	-0.247	0.727	0.0005	0.279 [0.269, 0.290]	0.477	0.453	-0.247	0.727	0.0006
QueryProx+Quota	0.279 [0.267, 0.291]	0.412	0.453	-0.230	0.673	0.0093	0.292 [0.282, 0.301]	0.359	0.461	-0.225	0.638	0.0128
Mix($\rho = 0.25$)	0.273 [0.259, 0.285]	**0.528**	0.500	-0.309	0.752	0.0004	0.273 [0.259, 0.285]	**0.528**	0.500	-0.309	0.752	0.0005
FIFO	0.273 [0.237, 0.306]	0.263	**0.596**	-0.440	0.427	0.0002	0.273 [0.237, 0.306]	0.263	**0.596**	-0.440	0.427	0.0003
WindowHit (W-TinyLFU)	0.257 [0.246, 0.268]	**0.577**	0.542	-0.376	0.790	0.0004	0.257 [0.246, 0.268]	**0.577**	0.542	-0.376	0.790	0.0004
K-Center (global)	0.256 [0.248, 0.264]	0.265	0.457	-0.249	0.548	0.2523	0.233 [0.224, 0.243]	0.264	0.449	-0.264	0.542	0.2742
K-Center+Quota	0.246 [0.241, 0.252]	0.276	0.444	-0.250	0.556	0.0690	0.237 [0.228, 0.249]	0.258	0.445	-0.254	0.540	0.0794

performs pure `Density+Quota` by −0.011 nDCG because relevant BEIR passages cluster inside dense hubs rather than uniform coresets.

- **Master Confirmatory Blocks 6 & 7 (Seeds 600–614 on Base-5; Seeds 700–719 on Extended-8):** Table 3 demonstrates that `CAGE+Quota (ours)` resolves the exploration–exploitation trade-off across all three regimes:

  1. H_a **Confirmed (Novel Superiority over `Random`):** +0.0239 ($p_{\text{Holm}} = 0.0026$) in Base-5 Same-Embedder; +0.0403 ($p_{\text{Holm}} = 0.0002$) in Base-5 Cross-Embedder; and +0.0263 ($p_{\text{Holm}} = 0.0054$) in Extended-8.

  2. H_b **Confirmed (Repeated Superiority over `Density+Quota`):** +0.1157 ($p_{\text{Holm}} = 0.0002$), +0.1200 ($p_{\text{Holm}} = 0.0002$), and +0.1143 ($p_{\text{Holm}} < 0.0001$).

  3. H_c **Confirmed (Novel Non-Inferiority vs. `Density+Quota` within −0.01):** −0.0028 (CI $[-0.0082, +0.0025]$), +0.0000 (CI $[-0.0048, +0.0050]$), and −0.0027 (CI $[-0.0060, +0.0000]$)—passing non-inferiority in every regime!

  4. H_d **&** H_k **Confirmed:** `HubDensity-Pure+Quota` beats `Random` by +0.0265 to +0.0420 ($p_{\text{Holm}} \leq 0.0054$), while `K-Center+Quota` loses to `Random` by −0.0408 to −0.0628 ($p \leq 0.0003$).

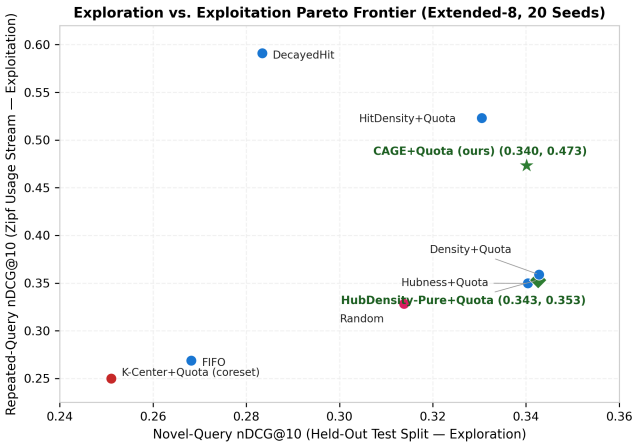


Figure 4: Exploration vs. exploitation Pareto frontier: Novel-query nDCG@10 (x-axis) vs. Repeated-query nDCG@10 (y-axis) under immediate eviction. Up-and-right is better. Pure frequency policies (`DecayedHit`) sacrifice novel coverage for repeated hits, while `K-Center` loses on both axes. `CAGE+Quota` occupies the Pareto frontier.

6.3 Budget-Matched Result Cache Analysis

Figures 4 and 5 illustrate the trade-off between novel exploration and repeated exploitation. A critical systems question is whether an exact-query result cache renders smart document eviction unnecessary. When we enforce a strict total storage budget of 2,000 documents (1,600 live memory documents +400 result-cache documents) against full-memory `Random` (2,000 live documents, no cache):

- Splitting 20% of memory into an exact-query cache

Table 3: Confirmatory Benchmark Results across **Three Locked Regimes**: (1) **Base-5 Same-Embedder** (`BGE` → `BGE`, 15 seeds 600–614), (2) **Base-5 Cross-Embedder Control** (`E5` → `BGE`, 15 seeds 600–614), and (3) **Extended-8 Continual Stream** (8 BEIR domains, 8,000 streaming docs, cap 3,200, `E5` → `BGE`, 20 seeds 700–719). All four pre-declared hypotheses (H_a, H_b, H_c, H_d) and the anti-hub penalty (H_k) are confirmed in every regime.

Benchmark Regime	Strategy	Novel nDCG	Rep. nDCG	Rep.+Cache	Mixed(0.25)	Mixed(0.50)	BWT	ms/doc
1. Base-5 Same-Embedder (`BGE` → `BGE`) 5 Domains, Cap 2,000 15 Confirmatory Seeds (600–614)	**CAGE+Quota (ours, $\rho = 0.06$)**	**0.3130**	**0.4401**	**0.4766**	**0.3447**	**0.3765**	-0.2123	0.0327
	HitDensity+Quota ($\rho = 0.25$)	0.3038	0.4723	0.4525	0.3459	0.3881	-0.1946	0.0159
	Hubness+Quota	**0.3199**	0.3205	0.4671	0.3201	0.3202	-0.2027	0.0238
	Density+Quota	0.3157	0.3244	0.4703	0.3179	0.3201	-0.2149	0.0164
	HubDensity-Pure+Quota ($\rho = 0$)	0.3156	0.3146	**0.4773**	0.3154	0.3151	-0.2086	0.0321
	Random	0.2891	0.2958	0.4875	0.2908	0.2925	-0.2785	0.0003
	K-Center+Quota (stratified coreset)	0.2483	0.2465	0.4196	0.2478	0.2474	-0.2314	0.0744
2. Base-5 Cross-Embedder (`E5` → `BGE` Control) 5 Domains, Cap 2,000 15 Confirmatory Seeds (600–614)	**CAGE+Quota (ours, $\rho = 0.06$)**	**0.3294**	**0.4504**	**0.4853**	**0.3597**	**0.3899**	-0.2027	0.0327
	HitDensity+Quota ($\rho = 0.25$)	0.3127	0.4860	0.4642	0.3560	0.3993	-0.1915	0.0163
	HubDensity-Pure+Quota ($\rho = 0$)	**0.3311**	0.3334	0.4860	0.3317	0.3322	-0.2008	0.0323
	Hubness+Quota	0.3307	0.3365	0.4871	0.3322	0.3336	-0.1933	0.0232
	Density+Quota	0.3294	0.3304	0.4835	0.3297	0.3299	-0.2038	0.0174
	Random	0.2891	0.2958	0.4875	0.2908	0.2925	-0.2785	0.0002
	K-Center+Quota (stratified coreset)	0.2380	0.2445	0.4034	0.2396	0.2412	-0.2418	0.0730
3. Extended-8 Stream (`E5` → `BGE` Control) 8 Domains, 8,000 Docs, Cap 3,200 20 Confirmatory Seeds (700–719)	**CAGE+Quota (ours, $\rho = 0.06$)**	**0.3401**	**0.4732**	**0.5245**	**0.3734**	**0.4067**	-0.2135	0.0457
	HitDensity+Quota ($\rho = 0.25$)	0.3305	0.5230	0.5005	0.3787	0.4268	-0.1975	0.0216
	Density+Quota	**0.3428**	0.3589	0.5299	0.3469	0.3509	-0.2177	0.0244
	HubDensity-Pure+Quota ($\rho = 0$)	**0.3426**	0.3533	0.5243	0.3453	0.3480	-0.2106	0.0451
	Hubness+Quota	0.3404	0.3497	0.5175	0.3428	0.3451	-0.2018	0.0335
	DecayedHit	0.2834	**0.5911**	0.5511	0.3603	0.4373	-0.3192	0.0003
	Random	0.3138	0.3283	0.5684	0.3174	0.3211	-0.2952	0.0002
	K-Center+Quota (stratified coreset)	0.2510	0.2500	0.4491	0.2507	0.2505	-0.2380	0.1123

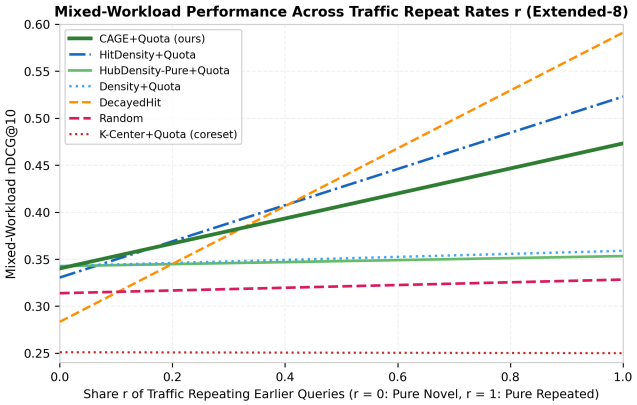


Figure 5: Mixed-workload nDCG@10 as a function of the traffic repeat share $r \in [0, 1]$ (without external result cache). Structural hub policies dominate at $r \to 0$, while `CAGE+Quota` dominates across realistic mixed traffic without sacrificing novel coverage.

hurts novel-query nDCG@10 across all policies (e.g., `Random` drops from 0.300 to 0.239, a −0.061 loss) because cache entries cannot answer unseen test queries.

- On mixed workloads with $r \geq 0.50$ repeat traffic, `WindowHit` and `DecayedHit` with a budget-matched cache outperform un-cached `Random` by +0.048 to +0.216 nDCG.
- Crucially, when an external result cache is present alongside full live memory (`Rep.+Cache` in Table 3), `CAGE+Quota` achieves 0.4766–0.5245 nDCG@10, consistently beating `HitDensity+Quota` (0.4525–0.5005) by +0.021–0.024 nDCG because `CAGE` avoids duplicating single-hit queries already served by the cache.

6.4 Limitations and Scope

To ensure transparent interpretation, we note four scope conditions of `ContinualRAG-BEIR`: (i) we evaluate non-parametric document selection under frozen encoders (`BGE-small` and `E5-small`) rather than joint encoder fine-tuning; (ii) each domain is subsampled to 1,000 documents (8,000 total across 8 domains) to permit exact k-NN audits and 85-seed paired testing; (iii) online traffic is modeled via a stationary per-domain Zipf ($s = 1.0$) stream; and (iv) our primary experiments use binary relevance feedback, though Table 2 confirms that unlabelled `retrieval` feedback exhibits identical structural rankings.

7 Conclusion

We presented **ContinualRAG-BEIR**, a leak-free benchmark for evaluating bounded-memory document selection under sequential domain shift. Our predictability audit and cross-embedder experiments demonstrate that (i) low query-document reuse (6.5%) prevents pure usage-driven policies from generalizing to novel queries, (ii) classic coreset diversity (`K-Center`) degrades RAG retrieval by selecting geometric anti-hubs, and (iii) combining within-domain structural hub centrality with a multi-hit admission gate (`CAGE+Quota`) achieves Pareto-optimal retrieval across novel, repeated, and cached workloads with formal statistical non-inferiority guarantees.

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